

KisanAI (SmartFarmAI)

AI Agent for Smart Farming

Objective

Develop an AI-powered Smart Farming Assistant that provides personalized, real-time agricultural guidance through crop recommendations, pest & disease support, RAG-based knowledge retrieval, multilingual AI chat, weather information, and mandi updates.

Problem Statement

Indian farmers often struggle to access timely and reliable information about crop selection, pest control, weather forecasts, and market prices. Information is scattered, difficult to understand, and often unavailable in local languages, resulting in poor decisions, crop losses, excessive pesticide use, and reduced income.

Proposed Solution

KisanAI is built using Python (Flask), IBM watsonx.ai, IBM Granite models, ChromaDB, Sentence-Transformers, and Retrieval-Augmented Generation (RAG). It offers an AI Chat Assistant, Crop Recommendation Engine, Pest & Disease Advisor, Weather & Mandi Information, and a simple mobile-friendly dashboard.

Technologies Used

- Python & Flask
- IBM watsonx.ai
- IBM Granite (ibm/granite-4-h-small)
- ChromaDB
- Sentence-Transformers
- Retrieval-Augmented Generation (RAG)
- HTML, CSS & JavaScript

Innovation & Uniqueness

Farmer-centric design, RAG-powered trustworthy responses, multilingual interaction, personalized recommendations, IBM Granite reasoning, and a lightweight mobile-friendly interface.

Future Scope

- Image-based pest detection
- Voice assistant in regional languages
- WhatsApp/SMS integration
- Smart weather alerts and notifications